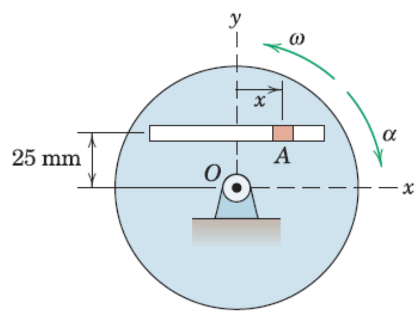
MEL1010 Engineering Mechanics
Tutorial 9: Motion Relative to rotating axes and Kinetics using Newtonian solution approach

Answer all of the following questions. Mention valid assumptions wherever applicable

<table>
<tr>
<td>1</td>
<td>The disk rotates about a fixed axis through O with angular velocity $\omega = 5$ rad/s and angular acceleration $\alpha = 3\,rad/s^2$ at the instant represented, in the directions shown. The slider A moves in the straight slot. Determine the absolute velocity and acceleration of A for the same instant, when $x = 36\,mm$, $\dot{x} = -100\,mm/s$, and $\ddot{x} = 150\,mm/s^2$.
Ans: $\boldsymbol{v}_A = -225\,\boldsymbol{i} + 180\boldsymbol{j}$ mm/s
$\boldsymbol{a}_A = -675\,\boldsymbol{i} - 1733\boldsymbol{j}$ mm/s^2</td>
<td>
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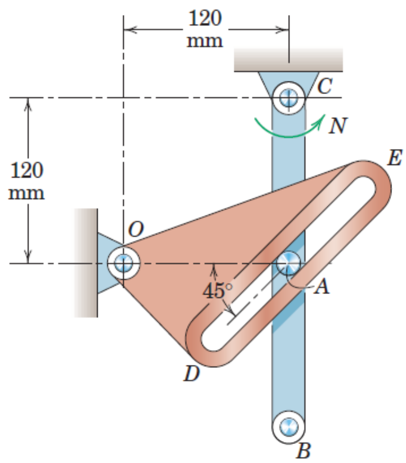
<td>2</td>
<td>For the instant represented, link CB is rotating counterclockwise at a constant rate $N = 4$ rad/s, and its pin A causes a clockwise rotation of the slot ted member ODE. Determine the angular velocity ω and angular acceleration α of ODE for this instant.
Ans. $\omega_{ODE} = 4$ rad/s CW, $\alpha_{ODE} = 64$rad/s^2 CCW</td>
<td>
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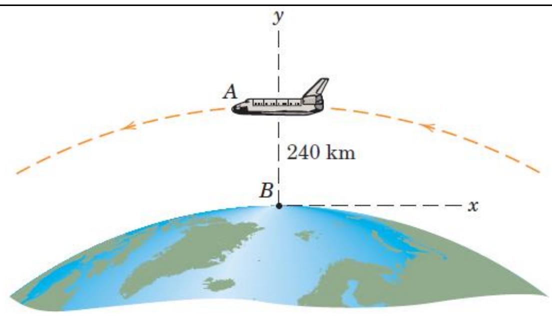
<td>3</td>
<td>The space shuttle A is in an equatorial circular orbit of $240 - km$ altitude and is moving from west to east. Determine the velocity and acceleration which it appears to have to an observer B fixed to and rotating with the earth at the equator as the shuttle passes overhead. Use $R = 6378$ km for the radius of the earth. Also use Fig. for the appropriate value of g and carry out your calculations to four-figure accuracy.
Ans: $\boldsymbol{v}_{rel} = -26220\,\boldsymbol{i}$ km/h
$\boldsymbol{a}_{rel} = -8.018\boldsymbol{j}$ m/s^2</td>
<td>
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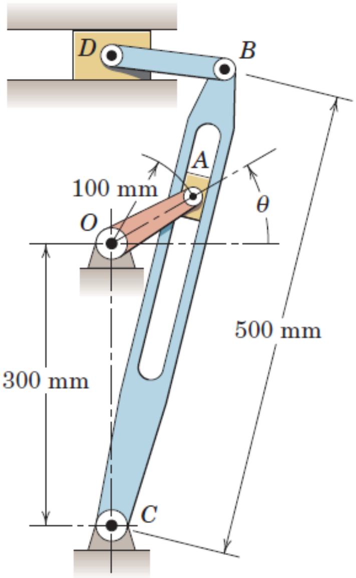
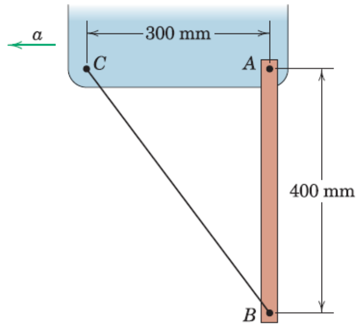
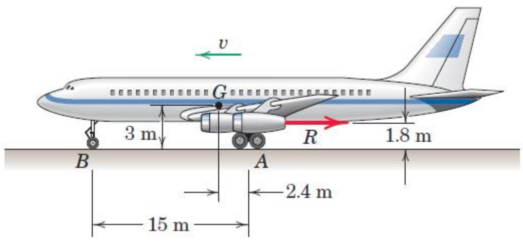
4	The figure illustrates a commonly used quick-return mechanism which produces a slow cutting stroke of the tool (attached to D) and a rapid return stroke. If the driving crank OA is turning at the constant rate $\dot{\theta} =$ 3 rad/s, determine the magnitude of the velocity of point B for the instant when $\theta =$ 30° Ans: v_B = 288 mm/s	
5	The uniform 5 kg bar AB is suspended in a vertical position from an accelerating vehicle and restrained by the wire BC. If the acceleration is a = 0.6 g, determine the tension T in the wire and the magnitude of the total force supported by the pin at A. Ans: T = 24.5 N, A = 32.9 N	
6	A jet transport with a landing speed of 200 km/h reduces its speed to 60 km/h with a negative thrust R from its jet thrust reversers in a distance of 425 m along the runway with constant deceleration. The total mass of the aircraft is 140 Mg with mass center at G. Compute the reaction N under the nose wheel B toward the end of the braking interval and prior to the application of mechanical braking. At the lower speed, aerodynamic forces on the aircraft are small and may be neglected. Ans: N = 257 N	

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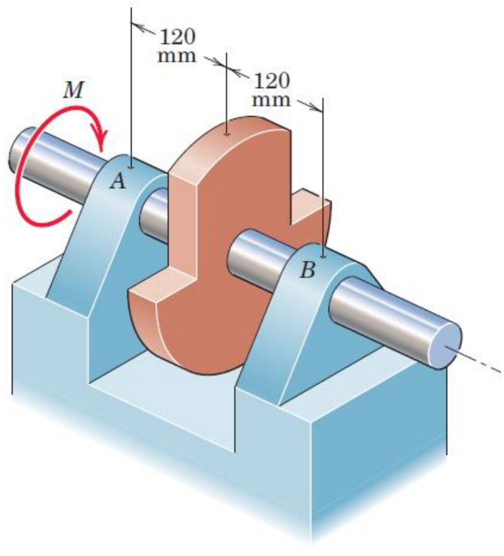
A vibration test is run to check the design adequacy of bearings A and B. The unbalanced rotor and attached shaft have a combined mass of 2.8 kg. To locate the mass center, a torque of 0.660 Nm is applied to the shaft to hold it in equilibrium in a position rotated 90° from that shown. A constant torque $M = 1.5\ N\ m$ is then applied to the shaft, which reaches a speed of 1200 rev/min in 18 revolutions starting from rest. During each revolution the angular acceleration varies, but its average value is the same as for constant acceleration. Determine (a) the radius of gyration k of the rotor and shaft about the rotation axis, (b) the force F which each bearing exerts on the shaft immediately after M is applied, and (c) the force R exerted by each bearing when the speed of 1200 rev/min is reached and M is removed. Neglect any frictional resistance and the bearing forces due to static equilibrium.

Ans: $k = 87.6$ mm, $F = 2.35$ N, R $= 531$ N



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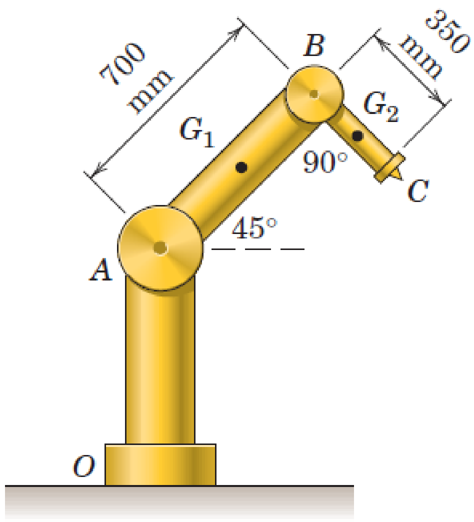
The robotic device consists of the stationary pedestal OA, arm AB pivoted at A, and arm BC pivoted at B. The rotation axes are normal to the plane of the figure. Estimate (a) the moment M_A, applied to arm AB required to rotate it about joint A at 4 rad/s^2 counter clockwise from the position shown with joint B locked and (b) the moment M_B applied to arm BC required to rotate it about joint B at the same rate with joint A locked. The mass of arm AB is 25 kg and that of BC is 4 kg, with the stationary portion of joint A excluded entirely and the mass of joint B divided equally between the two arms. Assume that the centers of mass G_1 and G_2 are in the geometric centers of the arms and model the arms as slender rods.

Ans: $M_A = 108.9$ N.m, $M_B = 5.51$ N.m



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Small ball bearing -rollers mounted on the ends of the slender bar of mass m and length l constrain the motion of the bar in the horizontal $x-y$ slots. If a couple M is applied to the bar initially at rest at $\theta = 45^{\circ}$, determine the forces exerted on the rollers at A and B as the bar starts to move.

Ans:

$$\boldsymbol{A} = \frac{3M}{2\sqrt{2}l}\boldsymbol{i}$$

$$\boldsymbol{B} = -\frac{3M}{2\sqrt{2}l}\boldsymbol{j}$$